D Definition T Theorem L Lemma C Corollary P Prop. Ex Example ! Idea/Remark

1 Fundamentals: Logic, sets, numbers and vectors

1.1 Real Numbers, Eulidean Spaces

Axioms

Axioms of addition:

$$\begin{array}{rll} \textbf{A1 (Assoc.)} & x+(y+z)=(x+y)+z & \forall x,y,z\in\mathbb{R} \\ \textbf{A2 (Neutral el.)} & x+0=x & \forall x\in\mathbb{R} \\ \textbf{A3 (Inverse el.)} & x+(-x)=0 & \forall x\in\mathbb{R}\ \exists(-x)\in\mathbb{R} \\ \textbf{A4 (Commut.)} & x+y=y+x & \forall x,y\in\mathbb{R} \end{array}$$

Axioms of Multiplication:

$$\begin{array}{rll} \textbf{M1 (Assoc.)} & x\cdot(y\cdot z)=(x\cdot y)\cdot z & \forall x,y,z\in\mathbb{R} \\ \textbf{M2 (Neutral el.)} & x\cdot 1=x & \forall x\in\mathbb{R} \\ \textbf{M3 (Inverse el.)} & x\cdot x^{-1}=1 & \forall x\in\mathbb{R}\setminus\{0\}\ \exists x^{-1}\in\mathbb{R} \\ \textbf{M4 (Commut.)} & x\cdot z=z\cdot x & \forall x,z\in\mathbb{R} \end{array}$$

Order Axioms:

$$\begin{array}{rll} \textbf{O1 (Reflexivity)} & x\le x & \forall x\in\mathbb{R} \\ \textbf{O2 (Transitivity)} & (x\le y\wedge y\le z)\Rightarrow x\le z & \\ \textbf{O3 (Antisymmetry)} & (x\le y\wedge y\le x)\Rightarrow x=y & \\ \textbf{O4 (Totality)} & \forall x,y\in\mathbb{R}:\ x\le y \text{ or } y\le x & \end{array}$$

Distributivity Axiom:

$$x\cdot(y+z)=xy+xz=(y+z)\cdot x \qquad \forall x,y,z\in\mathbb{R}$$

Compatibility Axioms:

$$\begin{array}{rll} \textbf{K1} & x\le y\Rightarrow x+z\le y+z & \forall x,y,z\in\mathbb{R} \\ \textbf{K2} & x\ge 0,\ y\ge 0\Rightarrow xy\ge 0 & \forall x,y\in\mathbb{R} \end{array}$$

R 1.1.5

The set $\mathbb{Q}$ of rational numbers, equipped with addition, multiplication, and the order relation $\le$, satisfies the above axioms.

V (Completeness axiom)

The set $\mathbb{R}$ is **order complete**: For any two nonempty sets $A, B \subset \mathbb{R}$ such that $a \le b$ for all $a \in A$, $b \in B$, there exists a number $c \in \mathbb{R}$ satisfying $a \le c \le b \quad \forall a \in A,\ b \in B$.

C 1.1.6

The consequences of above axioms are:

1. Uniqueness of the additive and multiplicative inverse.
2. $0 \cdot x = 0 \quad \forall x \in \mathbb{R}$.
3. $(-1)\cdot x = -x \quad \forall x \in \mathbb{R}$, in particular $(-1)^2 = 1$.
4. $y \ge 0 \iff (-y) \le 0$.
5. $y^2 \ge 0 \quad \forall y \in \mathbb{R}$, in particular $1 = 1\cdot 1 \ge 0$.
6. If $x \le y$ and $u \le v$, then $x+u \le y+v$.
7. If $0 \le x \le y$ and $0 \le u$, then $xu \le yu$.

T (Completeness of $\mathbb{R}$ – Dedekind completeness)

$\mathbb{R}$ is a commutative ordered field that is order-complete.

C 1.1.7 (Archimedean Principle)

1. For $x \in \mathbb{R}$ and $y > 0$ there exists $n \in \mathbb{N}$ such that $ny > x$.
- 2. For every $\varepsilon > 0$ there exists $n \in \mathbb{N}$ such that $\frac{1}{n} < \varepsilon$.

T 1.1.8 (Existence of Square Roots)

For every $t \ge 0$, $t \in \mathbb{R}$, the equation $x^2 = t$ has a solution in $\mathbb{R}$.

D (Bounded set - upper/lower bounded)

A set $A \in \mathbb{R}$ is called upper bounded if $\exists b \in R \ \ \forall a \in A : a \le b$. Each such b is called an upper bound for A. (Similarly for lower bound.)

T (Supremum/Infimum – Least Upper Bound Property)

1. Maximum and minimum (if exists) are unique
2. Every non-empty subset bounded below (above) has a unique infimum (supremum).
3. $\sup(X) \Leftrightarrow (\forall x \in X : x \le S) \wedge (\forall \varepsilon > 0\ \exists x \in X :\ x > S - \varepsilon)$ 4. $\inf(X) \Leftrightarrow (\forall x \in X : x \ge I) \wedge (\forall \varepsilon > 0\ \exists x \in X : x < I + \varepsilon)$

D (Maximum/Minimum)

$$\max\{x,y\} := \begin{cases} x, & \text{if } y \le x, \\ y, & \text{if } x \le y. \end{cases}, \ \min\{x,y\} := \begin{cases} y, & \text{if } y \le x, \\ x, & \text{if } x \le y. \end{cases}$$

T (Properties of the absolute value)

Properties:
1. $|x| \ge 0$.
2. $|x+y| \le |x| + |y|$ (triangle inequality).
3. $|x+y| \ge \big||x| - |y|\big|$ (inv. triangle inequality).

T (Young's inequality)

For every $\varepsilon > 0$ and all $x, y \in \mathbb{R}$ it holds: $2|xy| \le \varepsilon x^2 + \frac{1}{\varepsilon}y^2$.

1.2 Vectors and Vector Spaces

D (inner product)

For $x, y \in \mathbb{R}^n$ the standard inner product is $x \cdot y = \langle x, y\rangle = \sum_{i=1}^n x_i y_i$

D (Euclidean norm)

The Euclidean norm is $\|x\| = \sqrt{\sum_{i=1}^n x_i^2}$

D (Euclidean distance)

The Euclidean distance is $d(x,y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$

D (Triangle inequality)

For all $x, y, z \in \mathbb{R}^n$: $\|x - z\| \le \|x - y\| + \|y - z\|$

1.3 Complex Numbers

D (Complex number)

A complex number is $z = a + ib, \quad a, b \in \mathbb{R}$

D (set of complex numbers)

The set of complex numbers is $\mathbb{C} = \{a + ib \mid a, b \in \mathbb{R}\}$

D (Terminology of complex numbers)

Terminology
- $a = \mathrm{Re}(z)$ — real part
- $b = \mathrm{Im}(z)$ — imaginary part
- $i^2 = -1$
- If $a = 0$, the number is purely imaginary

D (Complex numbers operations)

Let $z_1 = x_1 + iy_1$, $z_2 = x_2 + iy_2$.
- **addition:** $z_1 + z_2 = (x_1 + x_2) + i(y_1 + y_2)$
- **subtraction:** $z_1 - z_2 = (x_1 - x_2) + i(y_1 - y_2)$
- **multiplication:** $z_1 z_2 = x_1 x_2 - y_1 y_2 + i(x_1 y_2 + y_1 x_2)$
- **division:** $\frac{z}{w} = \frac{z\overline{w}}{w\overline{w}} = \frac{z\overline{w}}{|w|^2}$
- **conjugate:** $\overline{z} = \overline{x + iy} = x - iy$

D (Properties of the Conjugate)

Properties are:
- $z\overline{z} = |z|^2$
- $z + \overline{z} = 2\,\mathrm{Re}(z), \quad z - \overline{z} = 2i\,\mathrm{Im}(z)$
- $\overline{\overline{z}} = z$
- $\overline{z \pm w} = \overline{z} \pm \overline{w}$
- $\overline{zw} = \overline{z}\,\overline{w}$
- $z = \overline{z}$, if $z \in \mathbb{R}$
- $\overline{z} = -z$, z is purely imaginary

D (Modulus)

The **magnitude** $|z|$ of a complex number $z = a + ib$ is the length of the corresponding vector: $|z| = \sqrt{a^2 + b^2}$
- The **distance** between two complex numbers: $d(z_1, z_2) = |z_2 - z_1|$
- **Triangle inequality**: $|z + w| \le |z| + |w|$

D (Euler's Formula)

The exponential function can be represented by a **series expansion**

$$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$$

If we substitute $x = it$ into this formula, we obtain **Euler's formula**.

$$e^{it} = \cos t + i \sin t$$

We can **rewrite** *sin(t)* and *cos(t)* using complex exponential functions:

$$\cos(t) = \frac{e^{it} + e^{-it}}{2}, \ \sin(t) = \frac{e^{it} - e^{-it}}{2i}$$

D (Exponential with Complex Argument)

We can also allow complex arguments for the exponential function. Then the following rules apply:

- $e^{z+w} = e^z e^w$
- $e^{a+ib} = e^a(\cos b + i \sin b)$
- $e^{z+i2\pi} = e^z$

D (Polar Form)

A complex number can also be described by specifying the distance from the origin and by specifying a suitable angle: $z = re^{i\varphi}$.

- $r = |z| \geq 0$ - distance
- $\varphi \in (-\pi, \pi]$ - polar angle
- $\varphi = \arg(z) = \operatorname{arc}(z)$

D (Polar to Cartesian conversion)

$\boxed{x = r\cos\varphi \mid y = r\sin\varphi}$.

D (Cartesian to Polar conversion)

$\boxed{r = \sqrt{x^2+y^2} \mid \varphi = \arctan \frac{y}{x}}$.

- If $x = 0$: $\arg(iy) = \begin{cases} \frac{\pi}{2}, & y > 0 \\ -\frac{\pi}{2}, & y < 0 \end{cases}$

D (Powers of Complex Numbers)

The following rules must be observed:

- $z_1 z_2 = r_1 r_2 e^{i(\varphi_1+\varphi_2)}$
- $\frac{z_1}{z_2} = \frac{r_1}{r_2} e^{i(\varphi_1-\varphi_2)}$
- $z^n = r^n e^{in\varphi}$

D (Roots from complex numbers)

For $n \in \mathbb{N}$, the solutions to the equation $z^n = re^{i\varphi}$ are

$$z_k = r^{1/n} e^{i(\varphi/n + 2\pi k/n)}, \quad k = 0, \ldots, n-1$$

D (Quadratic Equations)

For the quadratic equation $az^2 + bz + c = 0$, in complex numbers the solution is still $z_{1,2} = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$.

D (Fundamental Theorem of Algebra)

Each polynom $p(z) = \sum_{k=0}^{n} a_k z^k, \quad a_k \in \mathbb{C}$ can be factored into linear factors, i.e. written as $p(z) = a_n(z-z_1)(z-z_2)\ldots(z-z_n)$ The numbers z_k are therefore precisely the zero points of $p(z)$ (with **multiplicity**).

C (non-real roots appear in conjugate pairs)

Non-real roots of a polynomial with **real** coefficients occur in conjugate pairs $z, \overline{z}$.

2 Sequences

D (Sequence)

A sequence $(a_n)_{n\in\mathbb{N}_0}$ (also written $(a_n)_{n\geq 0}$ or $(a_n)_{n=0}^{\infty}$) is an infinite list of numbers, where each entry is called a *term* of the sequence. A sequence can also be viewed as a function $f : \mathbb{N}_0 \to \mathbb{R}$ (or $f : \mathbb{N} \to \mathbb{R}$), whose values are $f(n) = a_n$.

D (Ways to Describe a Sequence)

One way to describe a sequence is by giving an explicit formula for each term: $a_n = f(n)$. Another possibility is to define the sequence recursively, constructing each new term from previous ones: $a_n = g(a_{n-1}, a_{n-2}, \ldots)$. The functions f and g are called the *rule of formation* of the sequence.

D (Convergence/Divergence)

A sequence $(a_n)_{n\in\mathbb{N}_0} \subset \mathbb{R}$ is called *convergent* if there exists a number $L \in \mathbb{R}$ such that

$$\forall \varepsilon > 0 \; \exists N > 0 \text{ such that } \forall n > N : |a_n - L| < \varepsilon.$$

The number L is called the *limit* of the sequence, and we write

$$\lim_{n\to\infty} a_n = L.$$

If no such L exists, the sequence is called *divergent*.

T (Uniqueness of the Limit)

A convergent sequence has exactly one unique limit.

D (Divergence to infinity)

Let $(a_n)_{n\in\mathbb{N}_0}$ be a sequence.

- If $\forall M > 0 \; \exists N > 0$ such that $\forall n > N : a_n > M$, then the sequence diverges to infinity and we write $\lim_{n\to\infty} a_n = \infty$.
- If $\forall M > 0 \; \exists N > 0$ such that $\forall n > N : a_n < -M$, then the sequence diverges to minus infinity and we write $\lim_{n\to\infty} a_n = -\infty$.

D (Subsequence)

Let $(a_n)_{n\in\mathbb{N}_0}$ be a sequence in $\mathbb{R}$. A *subsequence* is a sequence of the form $(a_{n_k})_{k\in\mathbb{N}_0}$, where $(n_k)_{k\in\mathbb{N}_0}$ is a sequence of non-negative integers satisfying $n_{k+1} > n_k$ for all $k \in \mathbb{N}_0$.

L (Limit of a sequence and its subsequence)

Let $(a_n)_{n\in\mathbb{N}_0}$ be a convergent sequence with limit $L \in \mathbb{R}$. Then every subsequence $(a_{n_k})_{k\in\mathbb{N}_0}$ also converges to the same limit L.

D (Accumulation Point)

Let $(a_n)_{n\in\mathbb{N}_0}$ be a sequence in $\mathbb{R}$. A number $A \in \mathbb{R}$ is called an accumulation point of the sequence if

$$\forall \varepsilon > 0 \; \forall N \in \mathbb{N}_0 \; \exists n \geq N \text{ such that } |a_n - A| < \varepsilon.$$

Intuitively, every interval $(A-\varepsilon, A+\varepsilon)$ contains terms a_n for infinitely many indices n.

T (Characterization of Accumulation Points)

Let $(a_n)_{n\in\mathbb{N}_0}$ be a sequence in $\mathbb{R}$. A number $A \in \mathbb{R}$ is an accumulation point of the sequence if and only if there exists a convergent subsequence $(a_{n_k})_{k\in\mathbb{N}_0}$ such that $\lim_{k\to\infty} a_{n_k} = A$.

C (Accumulation Point of a Convergent Sequence)

Every convergent sequence $(a_n)_{n\in\mathbb{N}_0}$ in $\mathbb{R}$ has exactly one accumulation point, which coincides with its limit.

C (Infinite Occurrence Near an Accumulation Point)

Let A be an accumulation point of the sequence $(a_n)_{n\in\mathbb{N}_0}$. Then for every $\varepsilon > 0$ there exist infinitely many terms of the sequence lying in the interval $(A-\varepsilon, A+\varepsilon)$.

T (Algebra of Limits / Limit Laws)

Assume

$$\lim_{n\to\infty} a_n = K, \qquad \lim_{n\to\infty} b_n = L$$

with $K, L \neq \pm\infty$ and let C be a constant. Then

- $\lim_{n\to\infty}(a_n + b_n) = K + L$
- $\lim_{n\to\infty}(a_n - b_n) = K - L$
- $\lim_{n\to\infty}(a_n b_n) = KL$
- $\lim_{n\to\infty}(Ca_n) = CK$

If $L \neq 0$ and $b_n \neq 0$ then $\lim_{n\to\infty} \frac{a_n}{b_n} = \frac{K}{L}$.
If $K < L$ then there exists N such that $a_n < b_n$ for all $n \geq N$.
If there exists N such that $a_n \leq b_n$ for all $n \geq N$, then $K \leq L$.

T (Standard Limits)

More on the last page*

- $\lim_{n\to\infty} \frac{\log n}{n} = 0 = \lim_{n\to\infty} \frac{\ln n}{n}$
- $\lim_{n\to\infty} n^{1/n} = 1$
- $\lim_{n\to\infty} x^{1/n} = 1 \qquad (x > 0)$
- $\forall x \in \mathbb{R}: \quad \lim_{n\to\infty} \left(1 + \frac{x}{n}\right)^n = e^x$
- $\forall x \in \mathbb{R}: \quad \lim_{n\to\infty} \frac{x^n}{n!} = 0$

T (Squeeze Theorem / Sandwich Theorem)

Let $(a_n)_{n\in\mathbb{N}_0}$ and $(c_n)_{n\in\mathbb{N}_0}$ be convergent sequences with

$$\lim_{n\to\infty} a_n = L, \qquad \lim_{n\to\infty} c_n = L$$

and suppose there exists N_0 such that

$$a_n \leq b_n \leq c_n \quad \text{for all } n \geq N_0.$$

Then $(b_n)_{n\in\mathbb{N}_0}$ is also convergent and $\lim_{n\to\infty} b_n = L$.

D (Bounded Sequence)

A sequence $(a_n)_{n\in\mathbb{N}_0}$ is called bounded below or bounded above if the set $M = \{a_n \mid n \in \mathbb{N}_0\}$ has a lower bound or an upper bound respectively. A sequence that is bounded both above and below is called a bounded sequence.

D (Monotone Sequence)

A sequence $(a_n)_{n\in\mathbb{N}_0}$ is called:

- *monotone decreasing* if $m > n \Rightarrow a_m \leq a_n$.
- *monotone increasing* if $m > n \Rightarrow a_m \geq a_n$.
- *strictly monotone decreasing* if $m > n \Rightarrow a_m < a_n$.
- *strictly monotone increasing* if $m > n \Rightarrow a_m > a_n$.

T (Monotone Convergence Theorem / Weierstrass)

Every bounded and monotone sequence converges.

L (Convergent sequence = bounded)

Every convergent sequence is bounded.

T (Monotone Convergence)

Let $(a_n)_{n\in\mathbb{N}_0}$ be a sequence of real numbers.
- A monotone sequence converges if and only if it is bounded.
- (a_n) is monoton. increasing $\Rightarrow \lim_{n\to\infty} a_n = \sup\{a_n \mid n \in \mathbb{N}_0\}$.
- (a_n) is monoton. decreasing $\Rightarrow \lim_{n\to\infty} a_n = \inf\{a_n \mid n \in \mathbb{N}_0\}$.

D (Limit Superior and Limit Inferior)

Let $(a_n)_{n\in\mathbb{N}_0}$ be a sequence.
- *limit superior*: $\limsup_{n\to\infty} a_n = \lim_{n\to\infty} \sup\{a_k \mid k \geq n\}$
- *limit inferior*: $\liminf_{n\to\infty} a_n = \lim_{n\to\infty} \inf\{a_k \mid k \geq n\}$

L (Properties of limsup and liminf)

Let $(a_n)_{n\in\mathbb{N}_0}$ be a sequence. Then:

(i) $\limsup_{n\to\infty} a_n = \inf_{n\in\mathbb{N}} (\sup\{a_k \mid k \geq n\})$

(ii) $\liminf_{n\to\infty} a_n = \sup_{n\in\mathbb{N}} (\inf\{a_k \mid k \geq n\})$

(iii) If (a_n) is convergent, then

$$\lim_{n\to\infty} a_n = \limsup_{n\to\infty} a_n = \liminf_{n\to\infty} a_n.$$

(iv) A bounded sequence (a_n) converges if and only if

$$\lim_{n\to\infty} a_n = \limsup_{n\to\infty} a_n = \liminf_{n\to\infty} a_n.$$

T (Characterization via limsup)

Let $(a_n)_{n\in\mathbb{N}_0}$ be a bounded sequence and define $A = \limsup_{n\to\infty} a_n$. Then:

(i) A is an accumulation point of the sequence.

(ii) For every $\varepsilon > 0$, there are only finitely many indices n such that $a_n > A + \varepsilon$.

(iii) For every $\varepsilon > 0$, there are infinitely many indices n such that $A - \varepsilon < a_n < A + \varepsilon$.

An analogous statement holds for the limit inferior.

C (Bolzano–Weierstrass Theorem)

Every bounded sequence of real numbers has at least one accumulation point and admits a convergent subsequence.

D (Cauchy Sequence)

A sequence $(a_n)_{n\in\mathbb{N}_0} \subset \mathbb{R}$ is called a *Cauchy sequence* if

$$\forall \varepsilon > 0\ \exists N \in \mathbb{N} \text{ such that } \forall m, n \geq N: \ |a_n - a_m| < \varepsilon.$$

L (Cauchy Sequences are Bounded)

Every Cauchy sequence is bounded.

T (Cauchy Criterion)

A sequence of real numbers converges if and only if it is a Cauchy sequence.

3 Series

D (Series)

A (formal) expression of the form $a_0 + a_1 + a_2 + \cdots = \sum_{n=0}^{\infty} a_n$ is called a *series*. The terms a_n are called elements or summands. If the sequence of partial sums $s_n := \sum_{k=0}^{n} a_k$ converges to a limit $L \in \mathbb{R}$, then the series is said to *converge*, and we write $\sum_{n=0}^{\infty} a_n = L$. Otherwise, the series is called *divergent*.

T (Divergence Test / Necessary Condition for Convergence)

If the series $\sum_{n=0}^{\infty} a_n$ converges, then necessarily $\lim_{n\to\infty} a_n = 0$.

L (Linearity of Series)

Let $\sum_{n=0}^{\infty} a_n$ and $\sum_{n=0}^{\infty} b_n$ be convergent series. Then:

$$\sum_{n=0}^{\infty} (a_n + b_n) = \sum_{n=0}^{\infty} a_n + \sum_{n=0}^{\infty} b_n, \quad \sum_{n=0}^{\infty} C a_n = C \sum_{n=0}^{\infty} a_n \quad (C \in \mathbb{R}).$$

L (Tail of a Series)

Let $\sum_{n=0}^{\infty} a_n$ be a series. For any $N \in \mathbb{N}_0$: $\sum_{n=0}^{\infty} a_n$ converges $\Leftrightarrow$ $\sum_{n=N}^{\infty} a_n$ converges, and in that case

$$\sum_{n=0}^{\infty} a_n = \sum_{n=0}^{N-1} a_n + \sum_{n=N}^{\infty} a_n.$$

T (Monotone Convergence for Series / Non-Negative Terms)

Let $a_n \geq 0$ for all n. Then the sequence of partial sums $s_n = \sum_{k=0}^{n} a_k$ is monotonically increasing. If (s_n) is bounded, then the series $\sum_{n=0}^{\infty} a_n$ converges; otherwise, it diverges.

T (Comparison Test / Majorant–Minorant Criterion)

Let $\sum a_n, \sum b_n, \sum c_n$ be series with non-negative terms and suppose that for some N: $c_n \leq b_n \leq a_n$ for all $n \geq N$. Then:
- If $\sum a_n$ converges, then $\sum b_n$ converges (majorant test).
- If $\sum c_n$ diverges, then $\sum b_n$ diverges (minorant test).

D (Absolute and Conditional Convergence)

A series $\sum a_n$ is called *absolutely convergent* if $\sum |a_n|$ converges.
- It is called *conditionally convergent* if $\sum a_n$ converges but $\sum |a_n|$ does not.

T (Riemann Rearrangement Theorem)

Let $\sum a_n$ be a conditionally convergent series of real numbers, $L \in \mathbb{R}$. Then there exists a bijection $\varphi : \mathbb{N}_0 \to \mathbb{N}_0$ such that $\sum_{n=0}^{\infty} a_{\varphi(n)} = L$.

T (Dirichlet's Rearrangement Theorem)

Let $\sum a_n$ be absolutely convergent and let $\varphi : \mathbb{N}_0 \to \mathbb{N}_0$ be a bijection. Then $\sum_{n=0}^{\infty} a_{\varphi(n)}$ converges absolutely and $\sum_{n=0}^{\infty} a_n = \sum_{n=0}^{\infty} a_{\varphi(n)}$.

T (Leibniz Alternating Series Test)

Let (a_n) be a monotonically decreasing sequence of non-negative real numbers with

$$\lim_{n\to\infty} a_n = 0.$$

Then the alternating series

$$\sum_{n=0}^{\infty} (-1)^n a_n$$

converges. Moreover, for all $m \in \mathbb{N}_0$:

$$\sum_{n=0}^{2m+1} (-1)^n a_n \leq \sum_{n=0}^{\infty} (-1)^n a_n \leq \sum_{n=0}^{2m} (-1)^n a_n.$$

T (Cauchy Criterion for Series)

The series $\sum_{n=0}^{\infty} a_n$ converges if and only if for every $\varepsilon > 0$ there exists N such that for all $n \geq m \geq N$:

$$\left| \sum_{k=m+1}^{n} a_k \right| \leq \varepsilon.$$

T (Absolute Convergence $\Rightarrow$ Convergence)

Every absolutely convergent series converges, and

$$\left| \sum_{n=0}^{\infty} a_n \right| \leq \sum_{n=0}^{\infty} |a_n|.$$

T (Cauchy Root Test)

Let (a_n) be a sequence and define $\rho = \limsup_{n\to\infty} |a_n|^{1/n}$. Then:
- If $\rho < 1$, the series $\sum a_n$ converges absolutely.
- If $\rho > 1$, the series $\sum a_n$ diverges.

T (d'Alembert Ratio Test)

Let (a_n) with $a_n \neq 0$ and define $\rho = \lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right|$. Then:
- If $\rho < 1$, the series $\sum a_n$ converges absolutely.
- If $\rho > 1$, the series $\sum a_n$ diverges.

T (Cauchy Product / Mertens' Theorem)

Let $\sum a_n$ and $\sum b_n$ be absolutely convergent. Then

$$\left(\sum_{n=0}^{\infty} a_n \right) \left(\sum_{n=0}^{\infty} b_n \right) = \sum_{n=0}^{\infty} \left(\sum_{k=0}^{n} a_{n-k} b_k \right),$$

and the series on the right converges absolutely.

D (Power Series)

A series of the form $\sum_{k=0}^{\infty} c_k (x-a)^k$ is called a *power series* centered at a with coefficients c_k.

D (Convergence of a Power Series)

For fixed x, if the sequence of partial sums $s_n(x) = \sum_{k=0}^{n} c_k (x-a)^k$ converges, then the power series is said to converge at x.

T (Cauchy–Hadamard Theorem / Radius of Convergence)

Every power series has a radius of convergence R such that:
- The series converges for $|x-a|<R$,
- The series diverges for $|x-a|>R$.

The interval $(a-R, a+R)$ is called the interval of convergence. The radius R is given by $\rho = \limsup_{n\to\infty} |c_n|^{1/n}$, then:

$$R = \begin{cases} 0 & \text{if } \rho = \infty, \\ \frac{1}{\rho} & \text{if } 0 < \rho < \infty, \\ \infty & \text{if } \rho = 0. \end{cases}$$

4 Functions

4.1 Function basics

D Function

A function (mapping) is a rule that assigns to each admissible input exactly one output in the target set. The set of all possible inputs is called the *domain*, denoted by domain(f) or D. The set of all actually attained outputs is called the *image*, denoted by image(f) or W.

$$f : \text{domain}(f) \to \text{range}(f), \quad x \mapsto f(x).$$

The input is called the *independent variable*, the output the *dependent variable*.

D Restriction of a Function

Let $f : D(f) \to \mathbb{R}$ and let $D_0 \subseteq D(f)$. The restriction of f to D_0 is the function

$$f|_{D_0} : D_0 \to \mathbb{R}, \quad f|_{D_0}(x) = f(x).$$

Note that f and $f|_{D_0}$ are, in general, different functions.

D Bounded Functions

Let $f : D(f) \to \mathbb{R}$.
- f is *bounded above* if $\exists M \in \mathbb{R}$ such that $f(x) \le M \quad \forall x \in D(f)$.
- f is *bounded below* if $\exists M \in \mathbb{R}$ such that $f(x) \ge M \quad \forall x \in D(f)$.
- f is *bounded* if $\exists M \in \mathbb{R}$ such that $|f(x)| \le M \quad \forall x \in D(f)$.

D Injective, Surjective, Bijective

Let $f : X \to Y$.
- f is *injective* if $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$.
- f is *surjective* if $\forall y \in Y\ \exists x \in X : f(x) = y$.
- f is *bijective* if it is both injective and surjective.

D Composition

Let $f : X \to Y$ and $g : Y \to Z$. The composition is defined by

$$g \circ f : X \to Z, \quad (g \circ f)(x) = g(f(x)).$$

D Identity and Inverse

For a set X, the identity function is defined by $\mathrm{id}_X(x) = x$ $(\forall x \in X)$. If $f : X \to Y$ is bijective, then there exists a unique function $f^{-1} : Y \to X$ such that $f^{-1} \circ f = \mathrm{id}_X$, $f \circ f^{-1} = \mathrm{id}_Y$. This function is called the *inverse* of f.

D Monotonicity

Let $f : D(f) \subseteq \mathbb{R} \to \mathbb{R}$.
- f is *monotonically increasing* if $x_1 < x_2 \Rightarrow f(x_1) \le f(x_2)$.
- f is *strictly increasing* if $x_1 < x_2 \Rightarrow f(x_1) < f(x_2)$.
- f is *monotonically decreasing* if $x_1 < x_2 \Rightarrow f(x_1) \ge f(x_2)$.
- f is *strictly decreasing* if $x_1 < x_2 \Rightarrow f(x_1) > f(x_2)$.

T Strict Monotonicity Implies Injectivity

Every strictly monotone function is injective.

D Even and Odd Functions

Let $f : \mathbb{R} \to \mathbb{R}$.
- f is *even* if $f(-x) = f(x)$.
- f is *odd* if $f(-x) = -f(x)$.

D Convexity and Concavity (Geometric)

The graph of a function is called
- *convex* if it bends to the left when moving from left to right,
- *concave* if it bends to the right when moving from left to right.

4.2 Limits of Functions

D Limit of a Function (ε-δ definition)

Let $f : D(f) \to \mathbb{R}$ and $x_0 \in \mathbb{R}$. A number $L \in \mathbb{R}$ is called the limit of $f(x)$ at x_0 if

$$\forall \varepsilon > 0\ \exists \delta > 0 \text{ such that } \quad x \in D(f),\ |x - x_0| < \delta \Rightarrow |f(x) - L| < \varepsilon.$$

The limit is uniquely determined.

D One-Sided Limits

One-Sided Limits
- Right-hand limit: $\lim_{x\to x_0^+} f(x) = L$
- Left-hand limit: $\lim_{x\to x_0^-} f(x) = L$

D Limits at Infinity

Limits at Infinity
- $\lim_{x\to\infty} f(x) = L$
- $\lim_{x\to-\infty} f(x) = L$

D Infinite Limits

Infinite Limits
- $\lim_{x\to c} f(x) = \infty$
- $\lim_{x\to c} f(x) = -\infty$

T Limit Laws (Algebra of Limits for Functions)

Let

$$\lim_{x\to c} f(x) = K, \quad \lim_{x\to c} g(x) = L, \quad K, L \neq \pm\infty.$$

Then:
- $\lim_{x\to c}(f + g) = K + L$
- $\lim_{x\to c}(f - g) = K - L$
- $\lim_{x\to c}(fg) = KL$
- $\lim_{x\to c}(Af) = AK$
- If $L \neq 0$: $\lim_{x\to c} \frac{f}{g} = \frac{K}{L}$

4.3 Continuity of functions

D Continuity (ε-δ definition)

Let $f : D \to \mathbb{R}$ and $x_0 \in D$. The function f is called *continuous at* x_0 if

$$\forall \varepsilon > 0\ \exists \delta > 0 \text{ such that } |x - x_0| < \delta \ \Rightarrow\ |f(x) - f(x_0)| < \varepsilon.$$

The function is called *continuous* if it is continuous at every point in its domain.

D One-Sided Continuity

Let $f : D(f) \to \mathbb{R}$ and $x_0 \in D(f)$.
- f is *right-continuous* at x_0 if $\lim_{x\to x_0^+} f(x) = f(x_0)$.
- f is *left-continuous* at x_0 if $\lim_{x\to x_0^-} f(x) = f(x_0)$.

T Sequential Characterization of Continuity (Heine's Theorem)

Let $f : D \subseteq \mathbb{R} \to \mathbb{R}$ and $x_0 \in D$. Then f is continuous at x_0 if and only if for every sequence (x_n) with $x_n \to x_0$ we have $f(x_n) \to f(x_0)$.

T Algebra of Continuous Functions

Let $f, g : D \subseteq \mathbb{R} \to \mathbb{R}$ be continuous. Then:
- $f + g$ is continuous
- $f - g$ is continuous
- cf is continuous for any constant $c \in \mathbb{R}$
- $f \cdot g$ is continuous
- $\frac{f}{g}$ is continuous if $g \neq 0$

T Continuity of Composition

Let $f : I \to \text{image}(f)$ and $g : \text{image}(f) \to \mathbb{R}$ be continuous. Then the composition $g \circ f$ is continuous and

$$\lim_{x\to x_0} g(f(x)) = g\left(\lim_{x\to x_0} f(x)\right).$$

T Continuity of the Inverse Function

Let I be an interval and let $f : I \to J$ be continuous and bijective. Then J is also an interval and the inverse function $f^{-1} : J \to I$ is continuous.

T Intermediate Value Theorem

Let $f : [a, b] \to \mathbb{R}$ be continuous and let c be a value between $f(a)$ and $f(b)$. Then there exists $x \in [a, b]$ such that $f(x) = c$.

C Zero of a Continuous Function (Bolzano theorem)

Let $f : I \subseteq \mathbb{R} \to \mathbb{R}$ be continuous and suppose

$$f(a) < 0, \quad f(b) > 0.$$

Then there exists $x \in (a, b)$ such that

$$f(x) = 0.$$

D Compact Interval

A bounded and closed interval $[a, b]$ is called *compact*.

L Bolzano–Weierstrass Theorem (for Compact Intervals)

Let (x_n) be a sequence in a compact interval $[a, b]$. Then there exists a subsequence (x_{n_k}) such that $x_{n_k} \to x_0 \quad$ with $x_0 \in [a, b]$.

D Local Extrema

Let $f : D \subseteq \mathbb{R} \to \mathbb{R}$.

- x_0 is a *local maximum* if $f(x_0) \geq f(x) \quad \forall x \in D$.
- x_0 is a *local minimum* if $f(x_0) \leq f(x) \quad \forall x \in D$.
- Local maxima and minima are called *local extrema*.

T Extreme Value Theorem (Weierstrass)

Let $f : I \subseteq \mathbb{R} \to \mathbb{R}$ be continuous and let I be compact. Then f is bounded and attains its maximum and minimum on I, i.e., there exist $x_{\min}, x_{\max} \in I$ such that $f(x_{\min}) \leq f(x) \leq f(x_{\max})$ $(\forall x \in I)$.

5 Derivatives

5.1 Differentiation

D Derivative

Let $f : D \to \mathbb{R}$ and assume D has no isolated points. The *difference quotient* is

$$\frac{f(a+h) - f(a)}{h}.$$

If the limit exists, the *derivative* of f at a is defined by

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h} = \lim_{b \to a} \frac{f(b) - f(a)}{b - a}.$$

Interpretation:

- instantaneous rate of change
- slope of the tangent
- sign indicates increase/decrease

D Derivative Function

Given $f : D \to \mathbb{R}$, the *derivative function* is $x \mapsto f'(x)$.

D Differentiability

A function f is *differentiable at* x if $f'(x)$ exists.If this holds for all x in the domain, f is called differentiable.

L Differentiability Implies Continuity

If f is differentiable at x_0, then f is continuous at x_0.

D One-Sided Derivatives

Derivatives

- Right derivative: $\lim_{h \to 0^+} \frac{f(a+h)-f(a)}{h}$
- Left derivative: $\lim_{h \to 0^-} \frac{f(a+h)-f(a)}{h}$

D Higher Derivatives

Define iteratively:

$$f^{(0)} = f, \quad f^{(1)} = f', \quad f^{(2)} = f'', \quad f^{(n+1)} = (f^{(n)})'.$$

If $f^{(n)}$ exists, f is n-times differentiable. If all derivatives up to order n are continuous, then $f \in C^n(D)$. Define $C^\infty(D) := \bigcap_{n=0}^\infty C^n(D)$, called *smooth functions*.

T Basic Derivatives

Basic Derivatives, more in the end*

- Power: $(ax^p)' = apx^{p-1}$
- Exponential: $(a^x)' = \ln(a)\, a^x$
- Trig.: $(\sin x)' = \cos x, \quad (\cos x)' = -\sin x, \quad (\tan x)' = \frac{1}{\cos^2 x}$
- Hyperbolic: $(\sinh x)' = \cosh x, \quad (\cosh x)' = \sinh x$
- Inverse: $(\ln x)' = \frac{1}{x}, \quad (\arctan x)' = \frac{1}{1+x^2}$

T Derivative Rules

Derivative Rules

- **Sum and Product Rule.** For differentiable functions f and g,
$$(f+g)' = f' + g' \quad (fg)' = f'g + fg'.$$
- **Chain Rule.** If f is differentiable at x_0 and g is differentiable at $f(x_0)$, then
$$(g \circ f)'(x_0) = g'\big(f(x_0)\big)\, f'(x_0).$$
- **Quotient Rule.** If $g(x_0) \neq 0$, then
$$\left(\frac{f}{g}\right)'(x_0) = \frac{f'(x_0)g(x_0) - f(x_0)g'(x_0)}{g(x_0)^2}.$$
- **Derivative of the Inverse.** Let f be bijective and differentiable, with $f'(x_0) \neq 0$. Then
$$(f^{-1})'\big(f(x_0)\big) = \frac{1}{f'(x_0)}.$$

T Fermat's Theorem (Necessary Condition for Extremum)

If x_0 is a local extremum and f is differentiable at x_0, then $f'(x_0) = 0$.

T Characterization of Local Extrema

If x_0 is a local extremum, then at least one holds:

- x_0 is a boundary point
- f is not differentiable at x_0
- $f'(x_0) = 0$

T Rolle's Theorem

Let f be continuous on $[a, b]$ and differentiable on (a, b) with $f(a) = f(b)$.Then there exists $\xi \in (a, b)$ such that $f'(\xi) = 0$.

T Mean Value Theorem (Lagrange)

Let f be continuous on $[a, b]$ and differentiable on (a, b).Then there exists $\xi \in (a, b)$ such that

$$f'(\xi) = \frac{f(b) - f(a)}{b - a}.$$

T l'Hospital's Rule (0/0)

If $\lim_{x\to a} f(x) = \lim_{x \to a} g(x) = 0$ and $\lim_{x \to a} \frac{f'(x)}{g'(x)} = L$, then

$$\lim_{x \to a} \frac{f(x)}{g(x)} = L.$$

T l'Hospital's Rule (∞/∞)

If $\lim_{x\to a} |f(x)| = \lim_{x\to a} |g(x)| = \infty$ and $\lim_{x\to a} \frac{f'(x)}{g'(x)} = L$, then

$$\lim_{x \to a} \frac{f(x)}{g(x)} = L.$$

T l'Hospital at Infinity

If $\lim_{x \to \infty} f(x), g(x) \in \{0, \infty\}$ and $\lim_{x\to\infty} \frac{f'(x)}{g'(x)} = L$, then

$$\lim_{x \to \infty} \frac{f(x)}{g(x)} = L.$$

T Monotonicity Criterion (First Derivative Test)

$f'(x) \geq 0 \iff f$ is increasing

C Constant Functions

f is constant iff $f'(x) = 0 \quad \forall x$.

T Convexity via First Derivative

f is convex iff f' is increasing.

C Convexity via Second Derivative

If f is twice differentiable:

- $f''(x) \geq 0 \iff f$ is convex.
- $f''(x) \leq 0 \iff f$ is concave

T Second Derivative Test

Let f be twice differentiable near x_0 and suppose $f'(x_0) = 0$. Then: $f''(x_0) > 0 \quad \Rightarrow \quad x_0$ is a local minimum, and $f''(x_0) < 0 \quad \Rightarrow$ x_0 is a local maximum.

5.2 Taylor Approximation and Power Series

Local Approximation by Polynomials

In a sufficiently small neighborhood of a point x_0, the graph of a differentiable function f behaves almost like its tangent at x_0. Hence, the function can locally be approximated by its tangent.

D Linear Approximation / Tangent Approximation

If f is differentiable at x_0, then $f(x) \approx f(x_0) + f'(x_0)(x - x_0)$. The expression $P_1(x) = f(x_0) + f'(x_0)(x - x_0)$ is called the *linearization*, *tangent approximation*, or *first Taylor polynomial* of f at x_0.

T Taylor's Theorem (with Lagrange Remainder)

Let f be infinitely differentiable on an open interval I containing x_0. Then for every $x \in I$,

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \cdots + \frac{1}{n!} f^{(n)}(x_0)(x - x_0)^n,$$

for some c between x_0 and x. Equivalently,

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k + R_n(x),$$

where $R_n(x) = \frac{1}{(n+1)!} f^{(n+1)}(c)(x - x_0)^{n+1}$. The polynomial

$$P_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k$$

is called the *n-th Taylor polynomial.*

Taylor Approximation and Power Series

Taylor polynomials provide increasingly accurate approximations of many functions. For many functions, taking infinitely many terms recovers the original function.

D Taylor Series

For a function f, the series $\sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!}(x-a)^k$ is called the *Taylor series* (or Taylor expansion) of f around the point a.

T Important Taylor Series

The following series converge for all x:

$$\sin(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$$

$$\cos(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots$$

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

T Further Important Power Series (Geometric, Binomial)

The following series converge for $-1 < x < 1$.

$$\ln(1+x) = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^n}{n} = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots$$

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \cdots$$

$$(1+x)^p = \sum_{n=0}^{\infty} \binom{p}{n} x^n = 1 + px + \frac{p(p-1)}{2!}x^2 + \cdots$$

Operations on Taylor Series

New Taylor series can be constructed from known ones using substitution, multiplication, differentiation, integration.

T Cauchy Product of Power Series

Let

$$f(x) = \sum_{n=0}^{\infty} a_n x^n, \qquad g(x) = \sum_{n=0}^{\infty} b_n x^n,$$

with x inside both convergence intervals. Then

$$f(x)g(x) = \sum_{n=0}^{\infty} \left(\sum_{k=0}^{n} a_k b_{n-k} \right) x^n.$$

T Termwise Differentiation

Let $f(x) = \sum_{n=0}^{\infty} a_n x^n$ inside its convergence interval. Then f is differentiable and

$$f'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1}.$$

T Termwise Integration

Let $f(x) = \sum_{n=0}^{\infty} a_n x^n$ inside its convergence interval I. Then for every interval $[a,b] \subseteq I$,

$$\int_a^b f(x)\,dx = \sum_{n=0}^{\infty} \int_a^b a_n x^n \,dx.$$

6 Differential equations

D Differential Equation

A *differential equation* is an equation involving an unknown function together with its derivatives.

$$u'(t) = r(a - u(t))(b - u(t))$$

where r, a, b are constants, another example: $y^{(4)}(x) - 4y^{(3)}(x) + 5y''(x) - 4y'(x) + 4y(x) = 0$. The variables t, x are called *independent variables*, while u, y are called *dependent variables.*

D Linear Differential Equation

A differential equation of the form

$$a_n(x)y^{(n)}(x) + a_{n-1}(x)y^{(n-1)}(x) + \cdots + a_1(x)y'(x) + a_0(x)y(x) = s(x$$

is called a *linear differential equation* with coefficients $a_i(x)$.

- The *order* of the differential equation is the highest order derivative appearing in the equation. The function $s(x)$ is called the *forcing term*. If $s(x) = 0$, the equation is called *homogeneous*; otherwise it is called *inhomogeneous.*

D Differential Equation with Constant Coefficients

If all coefficient functions $a_i(x)$ are constant functions, the equation is called a *differential equation with constant coefficients.*

T Superposition Principle (Linearity of Solutions)

If $y_1(x)$ and $y_2(x)$ are solutions of a linear homogeneous differential equation, then $y(x) = C_1 y_1(x) + C_2 y_2(x)$ $(C_1, C_2 \in \mathbb{R})$, is also a solution of the same differential equation.

T Fundamental Solutions

A linear homogeneous differential equation of order n possesses n solutions $y_1(x), \ldots, y_n(x)$ such that the general solution is given by

$$y(x) = C_1 y_1(x) + \cdots + C_n y_n(x), \qquad C_1, \ldots, C_n \in \mathbb{R}.$$

These solutions are called *fundamental solutions.*

Characteristic Polynomial (Euler Ansatz)

Consider the differential equation

$$a_n u^{(n)}(x) + a_{n-1} u^{(n-1)}(x) + \cdots + a_1 u'(x) + a_0 u(x) = 0.$$

Using the ansatz $u(x) = e^{\lambda x}$ (Euler ansatz), we get the *characteristic polynomial*: $p(\lambda) = a_n \lambda^n + a_{n-1}\lambda^{n-1} + \cdots + a_1 \lambda + a_0$

T Solution of Linear Homogeneous Differential Equations

Consider the linear homogeneous differential equation with constant coefficients $a_n u^{(n)}(x) + a_{n-1} u^{(n-1)}(x) + \cdots + a_1 u'(x) + a_0 u(x) = 0$, $a_i \in \mathbb{R}$. Its characteristic polynomial is $a_n \lambda^n + a_{n-1}\lambda^{n-1} + \cdots + a_1 \lambda + a_0 = 0$. If all roots λ_i are real and have multiplicities m_i, then the general solution is

$$u(x) = \sum_{i=1}^{r} \left(\sum_{p=0}^{m_i - 1} C_{ip} x^p e^{\lambda_i x} \right).$$

If, in addition, there are complex conjugate roots $\alpha_j \pm i\beta_j$ with multiplicities m_j, then the corresponding real-valued part of the solution is

$$\sum_{j=1}^{s} \left(\sum_{q=0}^{m_j - 1} x^q e^{\alpha_j x} \left[A_{jq} \cos(\beta_j x) + B_{jq} \sin(\beta_j x) \right] \right).$$

Therefore, in the general case, the solution is

$$u(x) = \sum_{i=1}^{r} \left(\sum_{p=0}^{m_i - 1} C_{ip} x^p e^{\lambda_i x} \right) + \sum_{j=1}^{s} \left(\sum_{q=0}^{m_j - 1} x^q e^{\alpha_j x} \left[A_{jq} \cos(\beta_j x) + B_{jq}\, \mathrm{si} \right.\right.$$

Constants in the General Solution

The general solution of a differential equation of order n contains n constants. Therefore, in order to determine a unique solution, one must specify n additional conditions.

D Boundary Value Problem

Conditions imposed at different points, e.g. for $n = 2$: $u(t_0) = u_0, \quad u(t_1) = u_1, \quad t_0 \neq t_1$, define a *boundary value problem.*

D Initial Value Problem

Conditions imposed at a single point, e.g. for $n = 2$: $u(t_0) = u_0, \quad u'(t_0) = v_0$, define an *initial value problem.*

7 Integration Part 1

D Antiderivative / Indefinite Integral

Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$ be a function. A differentiable function $F : I \to \mathbb{R}$ satisfying $F'(x) = f(x) \quad \forall x \in I$ is called an *antiderivative* of f on I. Determining F from f is called *indefinite integration.*

Non-Uniqueness of Antiderivatives

An antiderivative is not unique. If $F(x)$ is an antiderivative of $f(x)$, then $F(x)+C$ is also an antiderivative for every constant $C \in \mathbb{R}$. Hence a function possesses an entire family of antiderivatives. This family is called the *indefinite integral* and is written as

$$\int f(x)\,dx.$$

The function f is called the *integrand*.

T Linearity of the Indefinite Integral

For indefinite integrals:

$$\int (f(x) \pm g(x))\,dx = \int f(x)\,dx \pm \int g(x)\,dx$$

and

$$\int s f(x)\,dx = s \int f(x)\,dx,$$

where s is a constant.

T Basic Integration Rules

Rules:

- $\int k\,dx = kx + C$
- $\int x^n\,dx = \frac{1}{n+1}x^{n+1} + C, \qquad n \neq -1$
- $\int e^x\,dx = e^x + C$
- $\int \frac{1}{x}\,dx = \ln(|x|) + C$
- $\int \sin(x)\,dx = -\cos(x) + C$
- $\int \cos(x)\,dx = \sin(x) + C$
- $\int \sinh(x)\,dx = \cosh(x) + C$
- $\int \cosh(x)\,dx = \sinh(x) + C$

T Integration by Parts

Using the product rule: $\frac{d}{dx}(f(x)g(x)) = f'(x)g(x) + f(x)g'(x)$, we obtain $\int \frac{d}{dx}(f(x)g(x))\,dx = (fg)(x)$. Therefore, $(fg)(x) = \int f'(x)g(x)\,dx + \int f(x)g'(x)\,dx$. Rearranging gives the formula for *integration by parts*:

$$\boxed{\int f'(x)g(x)\,dx = f(x)g(x) - \int f(x)g'(x)\,dx}$$

T Integration by Substitution

Using the chain rule: $\frac{d}{dx}f(g(x)) = f'(g(x))g'(x)$, we obtain $\int f'(g(x))g'(x)\,dx = f(g(x)) + C$. With the substitution $y = g(x)$, this becomes $\int f'(g(x))g'(x)\,dx = \int \frac{df}{dy}\,dy$. Hence the substitution rule is

$$\boxed{\int f'(g(x))g'(x)\,dx = f(g(x)) + C}$$

8 Integration Part 2

D Partition of an Interval

Let $[a,b] \subseteq \mathbb{R}$ be a compact interval. A *partition* of $[a,b]$ is a finite set of points

$$a = x_0 < x_1 < \cdots < x_{n-1} < x_n = b.$$

The points x_i are called *subdivision points*. A partition

$$a = y_0 < y_1 < \cdots < y_m = b$$

is called a *refinement* of $a = x_0 < x_1 < \cdots < x_n = b$ if $\{x_0, \ldots, x_n\} \subseteq \{y_0, \ldots, y_m\}$.

D Step Function

A function $f : [a,b] \to \mathbb{R}$ is called a *step function* if there exists a partition $a = x_0 < x_1 < \cdots < x_n = b$ such that for every $k = 1, \ldots, n$, the restriction $f|_{(x_{k-1}, x_k)}$ is constant.

D Integral of a Step Function

Let $f : [a,b] \to \mathbb{R}$ be a step function with respect to the partition $a = x_0 < x_1 < \cdots < x_n = b$. Then the integral of f over $[a,b]$ is defined by

$$\int_a^b f(x)\,dx := \sum_{k=1}^{n} c_k(x_k - x_{k-1}),$$

where c_k denotes the constant value of f on (x_{k-1}, x_k).

D Upper and Lower Sums

Let

$$f : [a,b] \to \mathbb{R}.$$

The set of upper sums is defined by

$$U(f) := \left\{ \int_a^b s(x)\,dx \;\middle|\; s \in TF,\ s \geq f \right\},$$

and the set of lower sums by

$$L(f) := \left\{ \int_a^b s(x)\,dx \;\middle|\; s \in TF,\ s \leq f \right\},$$

where TF denotes the set of step functions.

D Riemann Integral (Darboux Definition)

A bounded function $f : [a,b] \to \mathbb{R}$ is called *Riemann integrable* if

$$\sup L(f) = \inf U(f).$$

In this case, the common value is called the *Riemann integral* and is denoted by

$$\int_a^b f(x)\,dx := \sup L(f) = \inf U(f).$$

The numbers a, b are called the lower and upper limits of integration, and f is called the integrand.

T Properties of the Riemann Integral

Properties of the Riemann Integral:

(i) (Linearity) If $f, g : [a,b] \to \mathbb{R}$ are integrable and $\alpha, \beta \in \mathbb{R}$, then

$$\int_a^b (\alpha f + \beta g)(x)\,dx = \alpha \int_a^b f(x)\,dx + \beta \int_a^b g(x)\,dx.$$

(ii) (Monotonicity) If $f(x) \leq g(x)$ for all x, then

$$\int_a^b f(x)\,dx \leq \int_a^b g(x)\,dx.$$

(iii) (Triangle Inequality)

$$\left| \int_a^b f(x)\,dx \right| \leq \int_a^b |f(x)|\,dx.$$

(iv) (Reversal of Integration Limits)

$$\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx.$$

(v) (Splitting the Interval) For $a \leq c \leq b$:

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx.$$

T Monotone Functions are Integrable

Every monotone function $f : [a,b] \to \mathbb{R}$ is Riemann integrable.

T Continuous Functions are Integrable

Every continuous function $f : [a,b] \to \mathbb{R}$ is Riemann integrable.

D Pointwise Convergence of Functions

Let $(f_n)_{n \in \mathbb{N}_0}$ be a sequence of functions $f_n : D \to \mathbb{R}$, and let $f : D \to \mathbb{R}$. The sequence (f_n) converges *pointwise* to f if for every $x \in D$, the sequence $(f_n(x))_{n \in \mathbb{N}_0}$ converges to $f(x)$.

D Uniform Convergence

The sequence (f_n) converges *uniformly* to f if $\forall \varepsilon > 0\ \exists N$ such that for all $n \geq N$ and all $x \in D$, we have $|f_n(x) - f(x)| < \varepsilon$.

T Uniform Limit Theorem (Continuity)

Let $(f_n)_{n \in \mathbb{N}_0}$ be a sequence of continuous functions $f_n : D \to \mathbb{R}$ converging uniformly to $f : D \to \mathbb{R}$. Then f is continuous.

T Uniform Convergence and Integration (Interchange of Limit and Integral)

Let $(f_n)_{n \in \mathbb{N}_0}$ be a sequence of integrable functions $f_n : [a,b] \to \mathbb{R}$ converging uniformly to $f : [a,b] \to \mathbb{R}$. Then f is integrable and

$$\int_a^b f(x)\,dx = \lim_{n \to \infty} \int_a^b f_n(x)\,dx.$$

T Mean Value Theorem for Integrals

If $f : [a, b] \to \mathbb{R}$ is continuous, then there exists $c \in [a, b]$ such that $f(c) = \frac{1}{b-a} \int_a^b f(x)\,dx$. The quantity

$$\frac{1}{b-a} \int_a^b f(x)\,dx$$

is called the *average value* of f on $[a, b]$.

T Fundamental Theorem of Calculus

Let $f : [a, b] \to \mathbb{R}$ be continuous. Then for every constant $C \in \mathbb{R}$, the function

$$F(x) := \int_a^x f(y)\,dy + C$$

is an antiderivative of f. Moreover, every antiderivative of f is of this form.

C Integral Representation

Let $F : [a, b] \to \mathbb{R}$ be continuously differentiable. Then

$$F(x) = F(a) + \int_a^x F'(t)\,dt.$$

C Newton–Leibniz Formula

Let $f : [a, b] \to \mathbb{R}$ be continuous and let F be an antiderivative of f. Then

$$\int_a^b f(t)\,dt = F(b) - F(a).$$

T Integration by Parts

Let $f, g : [a, b] \to \mathbb{R}$ be continuously differentiable. Then

$$\int_a^b f(x)g'(x)\,dx = f(x)g(x)\Big|_a^b - \int_a^b f'(x)g(x)\,dx.$$

T Substitution Rule / Change of Variables (Version 1)

Let $f : I \to J$ be continuously differentiable and $g : J \to \mathbb{R}$ continuous. Then for every $[a, b] \subseteq I$,

$$\int_a^b g(f(x))f'(x)\,dx = \int_{f(a)}^{f(b)} g(y)\,dy.$$

T Substitution Rule (Version 2)

Let $f : I \to J$ be continuously differentiable and $g : J \to \mathbb{R}$ continuous. Assume $f'(x) \neq 0 \quad \forall x \in [a, b]$ and let $f^{-1} : [f(a), f(b)] \to \mathbb{R}$ denote the inverse of $f|_{[a,b]}$. Then

$$\int_a^b g(f(x))\,dx = \int_{f(a)}^{f(b)} g(y)(f^{-1})'(y)\,dy.$$

9 Differential equations - 2

D Separation of Variables

The method of separation of variables is particularly suitable for differential equations of the form $y'(x) = f(x) \cdot g(y)$. The idea is to move all terms involving the independent variable to one side and all terms involving the dependent variable to the other side, and then integrate both sides.

D Linear Substitution

A differential equation of the form

$$\frac{dy}{dx} = y' = f(ax + by + c), \qquad a, b, c \in \mathbb{R},$$

can be transformed into a differential equation with separable variables by the substitution: $u = ax + by + c$.

Autonomous Equation

After the substitution $u = ax + by + c$, the differential equation $\frac{dy}{dx} = f(ax + by + c)$ becomes

$$\frac{du}{dx} = a + b\frac{dy}{dx} = a + bf(u).$$

The independent variable x no longer appears explicitly on the right-hand side. Such a differential equation is called *autonomous*.

D Homogeneous Differential Equation

A differential equation of the form

$$\frac{dy}{dx} = g\Big(\frac{y}{x}\Big)$$

is called *homogeneous in the variables*. It can be transformed into a differential equation with separable variables by the substitution $u = \frac{y}{x}$.

D Associated Homogeneous Equation

Given an inhomogeneous differential equation

$$a_n(x)y^{(n)}(x) + \cdots + a_0(x)y(x) = s(x), \qquad s(x) \neq 0,$$

the equation $a_n(x)y^{(n)}(x) + \cdots + a_0(x)y(x) = 0$ is called the associated homogeneous differential equation.

T General Solution = Homogeneous + Particular (Superposition)

Let $y_p(x)$ be a particular solution of a linear inhomogeneous differential equation and let $y_h(x)$ be the general solution of the associated homogeneous equation.Then $y(x) = y_h(x) + y_p(x)$ is the general solution of the linear inhomogeneous differential equation.

D Variation of Constants

Consider a first-order linear inhomogeneous differential equation. A particular solution can be obtained by replacing the constant appearing in the general solution of the associated homogeneous equation by a function of the independent variable. This method is called *variation of constants*.

Ex. Variation of Constants Example

Given

$$y' - y = x,$$

the associated homogeneous equation

$$y' - y = 0$$

has general solution

$$y_h(x) = Ke^x.$$

The variation-of-constants ansatz is therefore

$$y_p(x) = K(x)e^x.$$

T Method of Undetermined Coefficients (Standard Ansätze)

For a linear inhomogeneous ODE $L(y) = s(t)$:

Forcing term $s(t)$	**Ansatz** $y_p(t)$
$a_0 + a_1t + \cdots + a_nt^n$	$C_0 + C_1t + \cdots + C_nt^n$
Ae^{kt}	Ce^{kt}
$A\sin(\omega t) + B\cos(\omega t)$	$C_1\sin(\omega t) + C_2\cos(\omega t)$

If the corresponding characteristic root has multiplicity m, multiply the ansatz by t^m. That is: $0 \rightsquigarrow t^m$, $k \rightsquigarrow t^m$, $i\omega \rightsquigarrow t^m$.

T Picard–Lindelöf Theorem (Existence & Uniqueness, First Order)

The initial value problem

$$y' = f(x, y), \qquad y(x_0) = y_0,$$

has, for sufficiently well-behaved functions f, a unique solution $x \mapsto y(x)$, $x \in I$, where the interval I may depend on x_0 and y_0.

T Picard–Lindelöf Theorem (Existence & Uniqueness, Higher Order)

The initial value problem

$$y^{(n)} + f_{n-1}(x)y^{(n-1)} + \cdots + f_1(x)y' + f_0(x)y = s(x),$$

with initial conditions

$$y(x_0) = y_0, \qquad y'(x_0) = v_0, \qquad \ldots, \qquad y^{(n-1)}(x_0) = z_0,$$

has, for sufficiently well-behaved functions $f_{n-1}, \ldots, f_0, s$, a unique solution $x \mapsto y(x)$, $x \in I$, where the interval I may depend on x_0 and the prescribed initial data.

10 Additional

Important Limits

- $\lim_{x\to\infty} \frac{1}{x} = 0$
- $\lim_{x\to\infty} \left(1 + \frac{1}{x}\right) = 1$
- $\lim_{x\to 0} \frac{\log(1-x)}{x} = -1$
- $\lim_{x\to 0} x \log x = 0$
- $\lim_{x\to\infty} e^x = \infty$
- $\lim_{x\to -\infty} e^x = 0$
- $\lim_{x\to 0} \frac{1-\cos x}{x^2} = \frac{1}{2}$
- $\lim_{x\to 0} \frac{e^x-1}{x} = 1$
- $\lim_{x\to\infty} e^{-x} = 0$
- $\lim_{x\to -\infty} e^{-x} = \infty$
- $\lim_{x\to 0} \frac{x}{\arctan x} = 1$
- $\lim_{x\to\infty} \arctan x = \frac{\pi}{2}$
- $\lim_{x\to\infty} \frac{e^x}{x^m} = \infty$
- $\lim_{x\to -\infty} x e^x = 0$
- $\lim_{x\to\infty} \left(\frac{x}{x+k}\right)^x = e^{-k}$
- $\lim_{x\to\infty} \ln x = \infty$
- $\lim_{x\to 0} \ln x = -\infty$
- $\lim_{x\to 0} \frac{a^x-1}{x} = \ln(a)$
- $\lim_{x\to 0} \frac{e^{ax}-1}{x} = a$
- $\lim_{x\to\infty} (1+x)^{1/x} = 1$
- $\lim_{x\to 0} (1+x)^{1/x} = e$
- $\lim_{x\to 0} \frac{\ln(1+x)}{x} = 1$
- $\lim_{x\to 1} \frac{\ln x}{x-1} = 1$
- $\lim_{x\to\infty} \frac{\ln x}{x} = 0$
- $\lim_{x\to\infty} \frac{\log x}{x^\alpha} = 0$
- $\lim_{x\to\infty} x^a q^x = 0,\ 0 \le q < 1$
- $\lim_{x\to\infty} n^{1/n} = 1$
- $\lim_{x\to\infty} \sqrt[x]{x} = 1$
- $\lim_{x\to\infty} \frac{2x}{2^x} = 0$
- $\lim_{x\to\pm\infty} \left(1 + \frac{1}{x}\right)^x = e$
- $\lim_{x\to\infty} \left(1 - \frac{1}{x}\right)^x = \frac{1}{e}$
- $\lim_{x\to\pm\infty} \left(1 + \frac{k}{x}\right)^{mx} = e^{km}$
- $\lim_{x\to 0} \frac{\sin x}{x} = 1$
- $\lim_{x\to\infty} \frac{\sin x}{x} = 0$
- $\lim_{x\to 0^+} x \ln x = 0$
- $\lim_{x\to(\pi/2)^-} \tan x = +\infty$
- $\lim_{x\to(\pi/2)^+} \tan x = -\infty$
- $\lim_{x\to 0} \frac{1}{\cos x} = 1$
- $\lim_{x\to 0} \frac{\cos x - 1}{x} = 0$
- $\lim_{x\to 0} \sqrt[n]{n!} = \infty$

Rules of derivation (holds only if f^{-1} exists and f continuous.)

- **Sum:** $(f+g)' = f' + g'$
- **Product:** $(fg)' = f'g + fg'$
- **Chain rule:** $(g \circ f)' = g'(f)f'$
- **Quotient rule:** $\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$
- **Derivative of the Inverse:** $(f^{-1})'(y_0) = \frac{1}{f'(x_0)}, \quad y_0 = f(x_0)$

Rules of integration

- **Linearity of Integration:** $\int (f+g) = \int f + \int g$ and $\int cf = c \int f$
- **Integration by Parts:** $\int f'g = fg - \int fg'$
- **Substitution:** $\int g(f(x)) f'(x)\, dx = \int g(y)\, dy$

Logarithms

- **Change of base:** $\log_a x = \frac{\ln x}{\ln a}$
- **Power rule:** $\log_a(x^y) = y \log_a x$
- **Product rule:** $\log_a(xy) = \log_a x + \log_a y$
- **Quotient rule:** $\log_a \left(\frac{x}{y}\right) = \log_a x - \log_a y$
- **Logarithm 1:** $\log_a(1) = 0\ \forall a \in \mathbb{N}$ ($x \le 0$ undefined)

Trigonometry

- **Tangent and cotangent:** $\tan x = \frac{\sin x}{\cos x}, \quad \cot x = \frac{\cos x}{\sin x}$
- **Pythagorean identity:** $\sin^2 x + \cos^2 x = 1$
- **Double-angle sine:** $\sin(2x) = 2 \sin x \cos x$
- **Double-angle cosine:** $\cos(2x) = \cos^2 x - \sin^2 x$
- **Angle addition sine:** $\sin(x+y) = \sin x \cos y + \cos x \sin y$
- **Angle addition cosine:** $\cos(x+y) = \cos x \cos y - \sin x \sin y$
- **Even/Odd symmetry:** $\cos(x) = \cos(-x)$ and $\sin(-x) = -\sin(x)$
- **Reflection at $\pi/2$:** $\cos(\pi - x) = -\cos(x)$ and $\sin(\pi - x) = \sin(x)$

Special Trigonometric Values

°	rad	$\sin(\xi)$	$\cos(\xi)$	$\tan(\xi)$
0°	0	0	1	0
30°	$\frac{\pi}{6}$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{3}}{3}$
45°	$\frac{\pi}{4}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	1
60°	$\frac{\pi}{3}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$
90°	$\frac{\pi}{2}$	1	0	$\varnothing$
120°	$\frac{2\pi}{3}$	$\frac{\sqrt{3}}{2}$	$-\frac{1}{2}$	$-\sqrt{3}$
135°	$\frac{3\pi}{4}$	$\frac{\sqrt{2}}{2}$	$-\frac{\sqrt{2}}{2}$	-1
150°	$\frac{5\pi}{6}$	$\frac{1}{2}$	$-\frac{\sqrt{3}}{2}$	$-\frac{\sqrt{3}}{3}$
180°	π	0	-1	0

Quadrant	Angle range	sin	cos	tan
I	0° to 90°	+	+	+
II	90° to 180°	+	−	−
III	180° to 270°	−	−	+
IV	270° to 360°	−	+	−

Hyperbolic

- **Hyperbolic sine:** $\sinh(x) := \frac{e^x - e^{-x}}{2} : \mathbb{R} \to \mathbb{R}$
- **Hyperbolic cosine:** $\cosh(x) := \frac{e^x + e^{-x}}{2} : \mathbb{R} \to [1, \infty)$
- **Hyperbolic tangent:** $\tanh(x) := \frac{\sinh(x)}{\cosh(x)} = \frac{e^x - e^{-x}}{e^x + e^{-x}} : \mathbb{R} \to (-1, 1)$
- **Inverse hyperbolic sine:** $\operatorname{arsinh} x = \ln(x + \sqrt{x^2 + 1})$
- **Inverse hyperbolic cosine:** $\operatorname{arcosh} x = \ln(x + \sqrt{x^2 - 1})$
- **Inverse hyperbolic tangent:** $\operatorname{artanh} x = \frac{1}{2} \ln \frac{1+x}{1-x}$

Complement trick

$$\sqrt{ax+b} - \sqrt{cx+d} = \frac{ax+b-(cx+d)}{\sqrt{ax+b} + \sqrt{cx+d}}$$

Derivatives and Antiderivatives

Antiderivative	Function	Derivative
$\frac{x^{n+1}}{n+1}$	x^n	nx^{n-1}
$\ln\lvert x\rvert$	$\frac{1}{x} = x^{-1}$	$-x^{-2} = -\frac{1}{x^2}$
$\frac{2}{3}x^{3/2}$	$\sqrt{x} = x^{1/2}$	$\frac{1}{2\sqrt{x}}$
$\frac{n}{n+1}x^{1/n+1}$	$\sqrt[n]{x} = x^{1/n}$	$\frac{1}{n}x^{1/n-1}$
e^x	e^x	e^x
$\exp(x)$	$\exp(x)$	$\exp(x)$
$\frac{1}{a(n+1)}(ax+b)^{n+1}$	$(ax+b)^n$	$n(ax+b)^{n-1}a$
$x(\ln\lvert x\rvert - 1)$	$\ln(x)$	$\frac{1}{x}$
$\frac{1}{\ln(a)}a^x$	a^x	$a^x \ln(a)$
$\frac{x}{\ln(a)}(\ln\lvert x\rvert - 1)$	$\log_a \lvert x\rvert$	$\frac{1}{x\ln(a)}$
$-\cos(x)$	$\sin(x)$	$\cos(x)$
$\sin(x)$	$\cos(x)$	$-\sin(x)$
$-\ln\lvert\cos(x)\rvert$	$\tan(x)$	$\frac{1}{\cos^2(x)}$
$\ln\lvert\sin(x)\rvert$	$\cot(x)$	$-\frac{1}{\sin^2(x)}$
$x\arcsin(x) + \sqrt{1-x^2}$	$\arcsin(x)$	$\frac{1}{\sqrt{1-x^2}}$
$x\arccos(x) - \sqrt{1-x^2}$	$\arccos(x)$	$-\frac{1}{\sqrt{1-x^2}}$
$x\arctan(x) - \frac{1}{2}\ln(x^2+1)$	$\arctan(x)$	$\frac{1}{x^2+1}$
$\ln\lvert\sin(x)\rvert$	$\cot(x)$	$-\frac{1}{\sin^2(x)}$
$\cosh(x)$	$\sinh(x)$	$\cosh(x)$
$\sinh(x)$	$\cosh(x)$	$\sinh(x)$
$\ln\lvert\cosh(x)\rvert$	$\tanh(x)$	$\frac{1}{\cosh^2(x)}$

Further Derivatives

$F(x)$	$f(x) = F'(x)$
$\frac{1}{a} \ln \lvert ax + b \rvert$	$\frac{1}{ax+b}$
$\frac{ax}{c} - \frac{ad-bc}{c^2} \ln \lvert cx + d \rvert$	$\frac{a(cx+d)-c(ax+b)}{(cx+d)^2}$
$\frac{x}{2}\sqrt{a^2+x^2} + \frac{a^2}{2} \ln \lvert x + \sqrt{a^2+x^2} \rvert$	$\sqrt{a^2+x^2}$
$\frac{x}{2}\sqrt{a^2-x^2} + \frac{a^2}{2} \arcsin\left(\frac{x}{\lvert a \rvert}\right)$	$\sqrt{a^2-x^2}$
$\frac{x}{2}\sqrt{x^2-a^2} - \frac{a^2}{2} \ln \lvert x + \sqrt{x^2-a^2} \rvert$	$\sqrt{x^2-a^2}$
$\ln \lvert x + \sqrt{x^2 \pm a^2} \rvert$	$\frac{1}{\sqrt{x^2 \pm a^2}}$
$\arcsin\left(\frac{x}{\lvert a \rvert}\right)$	$\frac{1}{\sqrt{a^2-x^2}}$
$\frac{1}{a} \arctan\left(\frac{x}{a}\right)$	$\frac{1}{a^2+x^2}$
$-\frac{1}{a}\cos(ax+b)$	$\sin(ax+b)$
$\frac{1}{a}\sin(ax+b)$	$\cos(ax+b)$
x^x	$x^x(1+\ln \lvert x \rvert)$
$(x^x)^x$	$(x^x)^x(x + 2x \ln \lvert x \rvert)$
x^{x^x}	$x^{x^x}(x^{x-1} + x^x(1+\ln \lvert x \rvert) \ln \lvert x \rvert)$
$\frac{1}{2}\left(x - \frac{1}{2}\sin(2x)\right)$	$\sin^2(x)$
$\frac{1}{2}\left(x + \frac{1}{2}\sin(2x)\right)$	$\cos^2(x)$
$\operatorname{arsinh}(x)$	$\frac{1}{\sqrt{1+x^2}}$
$\operatorname{arcosh}(x)$	$\frac{1}{\sqrt{x^2-1}}$
$\operatorname{artanh}(x)$	$\frac{1}{1-x^2}$